We began this chapter by reviewing the basics of ordinary supervised machine learning algorithms, such as image classifiers, and although recent successes in supervised learning are important and useful, supervised learning is not going to get us to artificial general intelligence (AGI). We ultimately seek general-purpose learning machines that can be applied to multiple problems with minimal to no supervision and whose repertoire of skills can be transferred across domains. Large data-rich companies can gainfully benefit from supervised approaches, but smaller companies and organizations may not have the resources to exploit the power of machine learning. General-purpose learning algorithms would level the playing field for everyone, and reinforcement learning is currently the most promising approach toward such algorithms.

RL research and applications are still maturing, but there have been many exciting developments in recent years. Google's DeepMind research group has showcased some impressive results and garnered international attention. The first was in 2013 with an algorithm that could play a spectrum of Atari games at superhuman levels. Previous attempts at creating agents to solve these games involved fine-tuning the underlying algorithms to understand the specific rules of the game, often called *feature engineering*. These feature engineering approaches can work well for a particular game, but they are unable to transfer any knowledge or skills to a new game or domain. DeepMind's deep Q-network (DQN) algorithm was robust enough to work on seven games without any game-specific tweaks (see figure 1.11). It had nothing more than the raw pixels from the screen as input and was merely told to maximize the score, yet the algorithm learned how to play beyond an expert human level.

Figure 1.11. DeepMind's DQN algorithm successfully learned how to play seven Atari games with only the raw pixels as input and the player's score as the objective to maximize. Previous algorithms, such as IBM's Deep Blue, needed to be fine-tuned to play a specific game.